Quantification of generalised measurement incompatibility of mutually unbiased bases

Raul-Andrei Pop

École Polytechnique

Advisors: Marc-Olivier Renou, Lucas-Amadeus Tendick
PhIQuS group, Inria Paris-Saclay / LIX

Lab Research Project, 2025–2026

Quantum states and projective measurements

A quantum system is described by a state vector $|\psi\rangle \in \mathcal{H}$ – or, more generally, a density operator $\rho \succeq 0$ with $\mathrm{Tr}(\rho) = 1$ – on a finite-dimensional Hilbert space $\mathcal{H} = \mathbb{C}^d$.

Projective measurement on $\mathcal{H}$: a family $\{\Pi_a\}_{a=1}^n$ of orthogonal projectors that sum to the identity,

$$\Pi_a \Pi_b \;=\; \delta_{ab}\, \Pi_a, \qquad \sum_a \Pi_a \;=\; I.$$

Born rule. The probability of obtaining outcome a on the state ρ is

$$P(a \mid \rho) \;=\; \mathrm{Tr}(\rho\, \Pi_a).$$

Outcomes are intrinsically random; non-commuting observables cannot be diagonalised in a common basis, hence cannot be read out by a single projective family.

POVMs: the most general quantum measurement

In practice (noisy detectors, indirect measurements, ancillary systems) one needs measurements that are not projective.

POVM – *positive operator-valued measure*: a family $\{M_a\}_{a=1}^n$ of Hermitian, positive semidefinite operators that sum to the identity,

$$M_a \succeq 0, \qquad \sum_a M_a = I.$$

Born rule extends component-wise: $P(a \mid \rho) = \mathrm{Tr}(\rho M_a)$.

- Every projective measurement is a POVM (with $M_a^2 = M_a$).
- POVMs allow $\mathrm{rank}(M_a) > 1$, more outcomes than $\dim \mathcal{H}$, and non-commuting elements.

POVMs are the natural setting for the joint-measurability question studied in this work.

Motivation: incompatibility as a resource

Classical: all observables can be measured jointly.
Quantum: some sets of measurements are incompatible – no single experiment reproduces their statistics.

Incompatibility is a necessary resource for

- Bell nonlocality, EPR steering, quantum contextuality
- Quantum key distribution, state discrimination, random access codes

Mutually unbiased bases (MUBs) are, in many scenarios, the *maximally incompatible* measurements: $|\langle b_i|b'_j\rangle|^2 = 1/d$ for every pair.

In this work

We compute the noise robustness of incompatibility for MUBs, and extend the framework to richer compatibility structures.

Setup: joint measurability & white-noise robustness

A family of POVMs $\{M_{a|x}\}$ is jointly measurable iff there exists a single parent POVM $\{G_\lambda\}$ and probabilities $p(a \mid x, \lambda)$ with

$$M_{a|x} = \sum_\lambda p(a \mid x, \lambda)\, G_\lambda .$$

Mix with white noise at level $\eta \in [0, 1]$: $M^\eta_{a|x} = \eta\, M_{a|x} + (1-\eta)\, \mathrm{Tr}(M_{a|x})\, I/d$.

Noise-robustness threshold

$$\eta^\star = \max\{\, \eta : \{M^\eta_{a|x}\} \text{ is jointly measurable} \,\}.$$

$\eta^\star$ is computable exactly by a semidefinite program (Wolf, Pérez-García, Fernández, 2009; Designolle *et al.*, PRL 122, 050402, 2019).

Why MUBs? – the most incompatible measurements

Two orthonormal bases $\{|b_i\rangle\}, \{|b'_j\rangle\}$ of $\mathbb{C}^d$ are mutually unbiased iff

$$\left|\langle b_i|b'_j\rangle\right|^2 = \frac{1}{d} \qquad \forall\, i, j,$$

i.e. preparing in any state of one basis produces a *uniformly random* outcome in the other.

Why are MUBs the natural target for studying incompatibility?

- Among all families of k POVMs in dimension d, MUBs are conjectured to minimise $\eta^\star$: they are the families that remain incompatible under the most white noise.
- Designolle *et al.* (2019) derived analytical bounds on $\eta^\star$ that are saturated specifically by MUB families in several regimes.
- For prime-power d, the Wootters–Fields construction (1989) gives an explicit complete set of $d+1$ MUBs; we work in $d \in \{2, 3, 5, 7, 11\}$.
- MUBs also appear in quantum tomography, QKD (e.g. BB84), and entropic uncertainty relations.

$\Rightarrow$ MUBs are the canonical benchmark family for any new incompatibility scenario.

Compatibility hypergraphs (Quintino *et al.*, 2019)

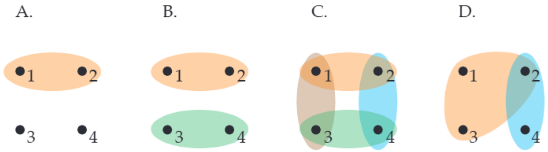
For 3 or more measurements, compatibility can be studied in a more complex way: for example triples can be pairwise compatible yet not jointly measurable as a whole.

We can model a set of measurements as a hypergraph $\mathcal{C} = [C_1, \ldots, C_N]$, where each hyperedge lists a subset that must be jointly measurable.

Our contribution. An SDP parameterised by an *arbitrary* hypergraph. Two canonical cases:

`compatible(k)` $\mathcal{C} = \{(1, \ldots, k)\}$
$\Rightarrow$ full k-wise JM.

`pair(k)` $\mathcal{C} = \{(i,j) : i < j\}$
$\Rightarrow$ only jointly measure pairs (i,j)



Examples on 4 measurements (Fig. 1, arXiv:1902.05841).

Pairwise generalization + closed-form lower bound

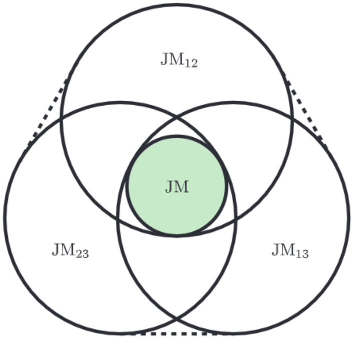
Geometric picture. Pairwise-compatible families lie in the convex hull of single-pair JM families. Strictly weaker than full JM.

Equal-weight case: $p_{ij} = 1/\binom{k}{2}$. Branch-counting under MUB symmetry yields:

Closed-form bound for pairwise generalization

$$\eta^{\star}_{(k-1)}(k,d) \;=\; 1 \;-\; \frac{\sqrt{d}}{k\,(\sqrt{d}+1)}.$$

Tight to solver precision in every case we checked.



$JM \subset \mathrm{conv}(JM_{\mathbf{12}}, JM_{\mathbf{13}}, JM_{\mathbf{23}})$.

Numerical results

Table 1. Full JM (reproduces Designolle et. al., 2019)

$k \setminus d$	2	3	5	7
2	0.7071	0.6830	0.6545	0.6371
3	0.5774	0.5686	0.5393	0.5101
4	—	0.5000	0.4615	0.4436
5	—	—	X	X
6	—	—	0.4082	X
8	—	—	—	0.3536

Matches closed forms to 10^{-4}.
"X" = computationally expensive (hours for $d^k > 3000$).

Table 2. All-pairs hypergraph (new)

$k \setminus d$	2	3	5	7	11
2	0.7071	0.6830	0.6545	0.6371	0.6158
3	0.8047	0.7887	0.7697	0.7581	0.7439
4	—	0.8415	0.8273	0.8186	0.8079
5	—	—	0.8618	0.8549	0.8463
6	—	—	0.8848	0.8791	0.8719
7	—	—	—	0.8963	0.8902
8	—	—	—	0.9093	0.9040
9	—	—	—	—	0.9146

Every entry matches the analytical bound
$\Rightarrow$ tightness of the equal-weight mixture.

Qualitative observation: robustness increases with k in the pairwise case

	Full JM $\eta^\star_{\mathrm{JM}}(k,d)$	**Pairwise Generalization** $\eta^\star_{(k-1)}(k,d)$
Behaviour in k	decreasing	increasing
Limit as $k \to \infty$	$\to 0$	$\to 1$
Limit as $d \to \infty$	$\to 0$	$\to 1 - 1/k$

- In the pairwise SDP, a fixed measurement appears in only $k-1$ of the $\binom{k}{2}$ branches – a vanishing fraction $2/k$. Its "responsibility" dilutes as k grows.
- Explicit instance of the Quintino et al. hierarchy: finer hypergraphs $\Rightarrow$ looser constraint $\Rightarrow$ larger $\eta^\star$.

Summary and outlook

Contributions.

1. Reproduced the standard JM noise-robustness SDP for MUBs in prime dimensions $d \in \{2, 3, 5, 7\}$.
2. Generalised the program to arbitrary compatibility hypergraphs.
3. Produced numerical results for the all-pairs hypergraph
4. Derived the closed-form bound $\eta^\star = 1 - \sqrt{d}/[\,k(\sqrt{d} + 1)\,]$ for the generalized pairwise incompatibility, numerically tight in every case.

Three natural extensions.

(i) *Dual SDP* $\Rightarrow$ analytical proof of tightness.

(ii) *Intermediate hypergraphs* (triples, $(k-2)$-subsets).

(iii) *Composite dimensions* ($d = 6$: open MUB problem).

Thank you!

Questions

Questions – the qubit SDP, explicitly

Pauli observables and their eigenprojectors:

$$\sigma_X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \Pi^{(X)}_{\pm} = \tfrac{1}{2}(I \pm \sigma_X), \quad \Pi^{(Z)}_{\pm} = \tfrac{1}{2}(I \pm \sigma_Z).$$

Noisy POVM elements $M^{\eta}_{a|x} = \eta\, \Pi^{(x)}_a + (1-\eta)\, I/2$:

$$M^{\eta}_{+|X} = \tfrac{1}{2}\begin{pmatrix} 1 & \eta \\ \eta & 1 \end{pmatrix}, \quad M^{\eta}_{-|X} = \tfrac{1}{2}\begin{pmatrix} 1 & -\eta \\ -\eta & 1 \end{pmatrix}, \quad M^{\eta}_{+|Z} = \tfrac{1}{2}\begin{pmatrix} 1+\eta & 0 \\ 0 & 1-\eta \end{pmatrix}, \quad M^{\eta}_{-|Z} = \tfrac{1}{2}\begin{pmatrix} 1-\eta & 0 \\ 0 & 1+\eta \end{pmatrix}.$$

Find $G_{\lambda_X \lambda_Z} \in \mathbb{C}^{2\times 2}$ Hermitian:

$$\begin{aligned} \text{maximise} \quad & \eta \in [0,1] \\ \text{s.t.} \quad & G_{\lambda_X\lambda_Z} \succeq 0, \ \textstyle\sum_{\lambda_X,\lambda_Z} G_{\lambda_X\lambda_Z} = I, \ \sum_{\lambda_Z} G_{a\,\lambda_Z} = M^{\eta}_{a|X}, \ \sum_{\lambda_X} G_{\lambda_X\,a} = M^{\eta}_{a|Z}. \end{aligned}$$

Optimal value

$\eta^{\star}(2,2) = 1/\sqrt{2} \approx 0.7071$ — matches our $(k,d) = (2,2)$ entry.

Questions – the general k-MUB joint-measurability SDP

For k noisy MUBs in dimension d, with $m = d$ outcomes per setting:

$$\begin{aligned} \text{maximise} \quad & \eta \in [0,1] \\ \text{s.t.} \quad & G_{\lambda_1 \cdots \lambda_k} \succeq 0 \quad \forall (\lambda_1, \ldots, \lambda_k) \in [\![1, m]\!]^k, \\ & \textstyle\sum_{\lambda_1, \ldots, \lambda_k} G_{\lambda_1 \cdots \lambda_k} = I, \\ & \textstyle\sum_{\lambda : \lambda_j = a} G_{\lambda_1 \cdots \lambda_k} = M^{\eta}_{a|j} \quad \forall j \in [\![1, k]\!], \ \forall a \in [\![1, m]\!]. \end{aligned}$$

Variable count: d^k Hermitian $d \times d$ matrices – exponential in k.

Questions – why pairwise scales well but full JM does not

Per-cell SDP variable count:

(k,d)	full JM d^k	all-pairs $\binom{k}{2}(d^2+(k-2)d)$	ratio
$(4,5)$	625	210	3×
$(4,7)$	2,401	378	6×
$(5,7)$	16,807	630	27×
$(6,7)$	117,649	945	124×
$(8,7)$	5,764,801	2,548	2,262×

Full JM: $\mathcal{O}(d^k)$ (exponential). Pairwise: $\mathcal{O}(k^3 d^2)$ (polynomial).

Why hours? Each interior-point iteration costs $\mathcal{O}(d^k \cdot d^3) \approx 2 \cdot 10^9$ operations just for the per-block eigendecompositions, with hundreds of iterations to converge – plus a multi-GB memory footprint that defeats cache. CVXPY canonicalisation alone takes tens of minutes at this size.

Hence Table 1 has many empty cells; Table 2 is complete.

Questions – derivation of the closed-form bound

Equal-weight case $p_{ij} = 1/\binom{k}{2}$. By MUB symmetry, every setting x is presented in $k-1$ *active* branches at noise $\eta^\star(2,d)$ and in $\binom{k}{2} - (k-1)$ *inactive* branches at noise 1.

Affineness of $\eta \mapsto M^\eta$:

$$\eta_{\text{eff}} \;=\; \frac{1}{\binom{k}{2}}\Big[\,(k-1)\,\eta^\star(2,d) + \frac{(k-1)(k-2)}{2}\,\Big] \;=\; \frac{1}{k}\big[\,2\,\eta^\star(2,d) + k - 2\,\big].$$

Substituting $\eta^\star(2,d) = (2+\sqrt{d})/[\,2(\sqrt{d}+1)\,]$ (Designolle):

$$\eta_{\text{eff}} \;=\; 1 + \frac{1}{k}\left(\frac{2+\sqrt{d}}{\sqrt{d}+1} - 2\right) \;=\; 1 - \frac{\sqrt{d}}{k\,(\sqrt{d}+1)}.$$

Boundary check: $k = 2$ recovers $\eta^\star(2,d) = (2+\sqrt{d})/[\,2(\sqrt{d}+1)\,]$. ✓

Questions – implementation & code

All SDPs formulated with CVXPY (Python).

- Solver: Clarabel for small cases ($d^k \leq 100$), Scs for the rest ($\epsilon = 10^{-7}$, 10^5 iterations).
- Cap: $d^k > 3000$ omitted in Table 1; no cap for Table 2 (polynomial scaling).
- Every closed-form entry of [?] reproduced to 10^{-4}; solver status `optimal` in every reported cell.

Core routines (`compatibility_hypergraph.py`):

- `robustness_hypergraph(M, hypergraph)` – the generalised SDP driver.
- `hypergraph_compatible(k)` – $\{(1, \ldots, k)\}$ for full JM.
- `hypergraph_pair(k)` – $\{(i,j) : i < j\}$ for pairwise generalization.